



Engineering Project Management & Finance **(UESTC 3031 & UESTCHN 3012)**

Project Management **(Part 2)**

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Outline



- **Part 1:** Introduction, The Role of the Project Manager
- **Part 2:** Project Scope Management, Project Schedule Management
- **Part 3:** Project Cost Management, Project Resource Management
- **Part 4:** Project Communications Management, Project Risk Management

PROJECT SCOPE MANAGEMENT



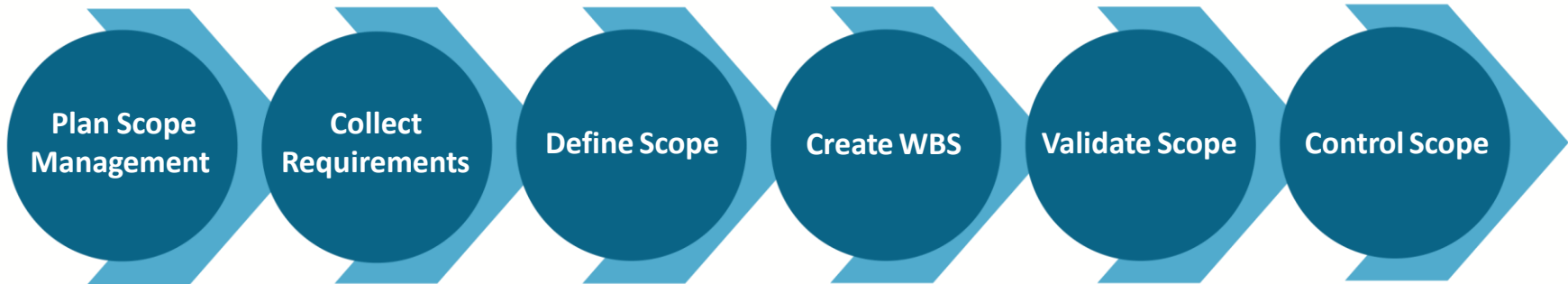
Session Overview

- Project Scope Management
- Product Scope and Project Scope
- Scope Management Processes
 - Plan Scope Management
 - Collect Requirements
 - Define Scope
 - Create WBS
 - Validate Scope
 - Control Scope

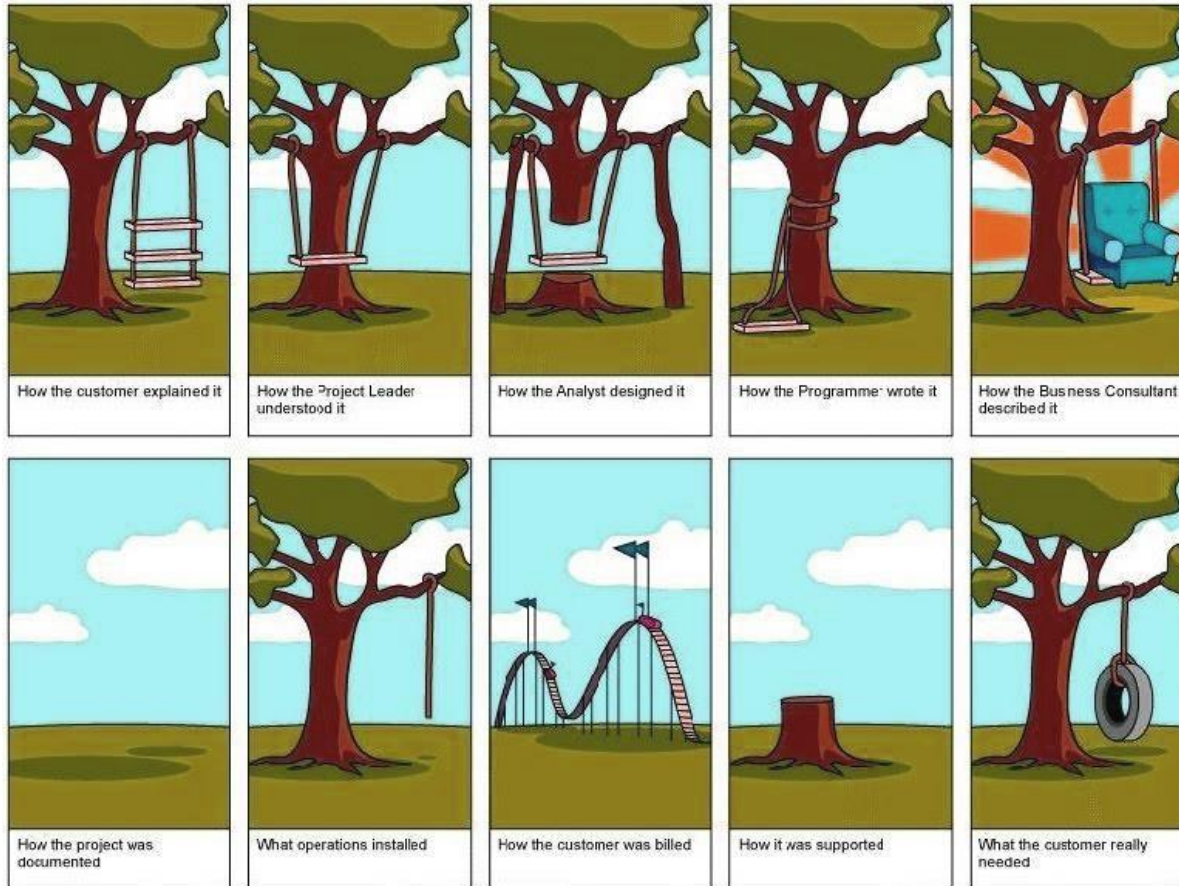


Processes in Scope management

- **Project Scope Management** includes the processes required to ensure that the project **includes all the work required, and only the work required**, to complete the project successfully



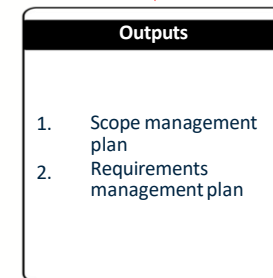
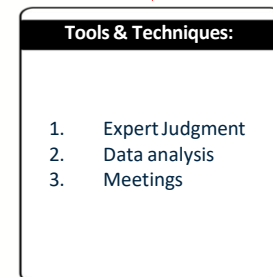
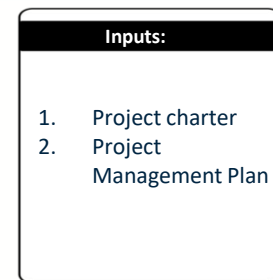
Project Scope Management



Plan Scope Management



- The process of creating a scope management plan that documents **how the project and product scope will be defined, validated, and controlled**
- The key benefit of this process is that it provides guidance and direction on how scope will be managed throughout the project



Collect Requirements



- The process of **determining, documenting, and managing stakeholder needs and requirements** to meet project objectives
- The key benefit of this process is that it provides the basis for defining the product scope and project scope



Inputs:

1. Project charter
2. Project management plan
3. Project documents
4. Business documents
5. Agreements
6. Enterprise environmental factors
7. Organizational process assets



Tools & Techniques:

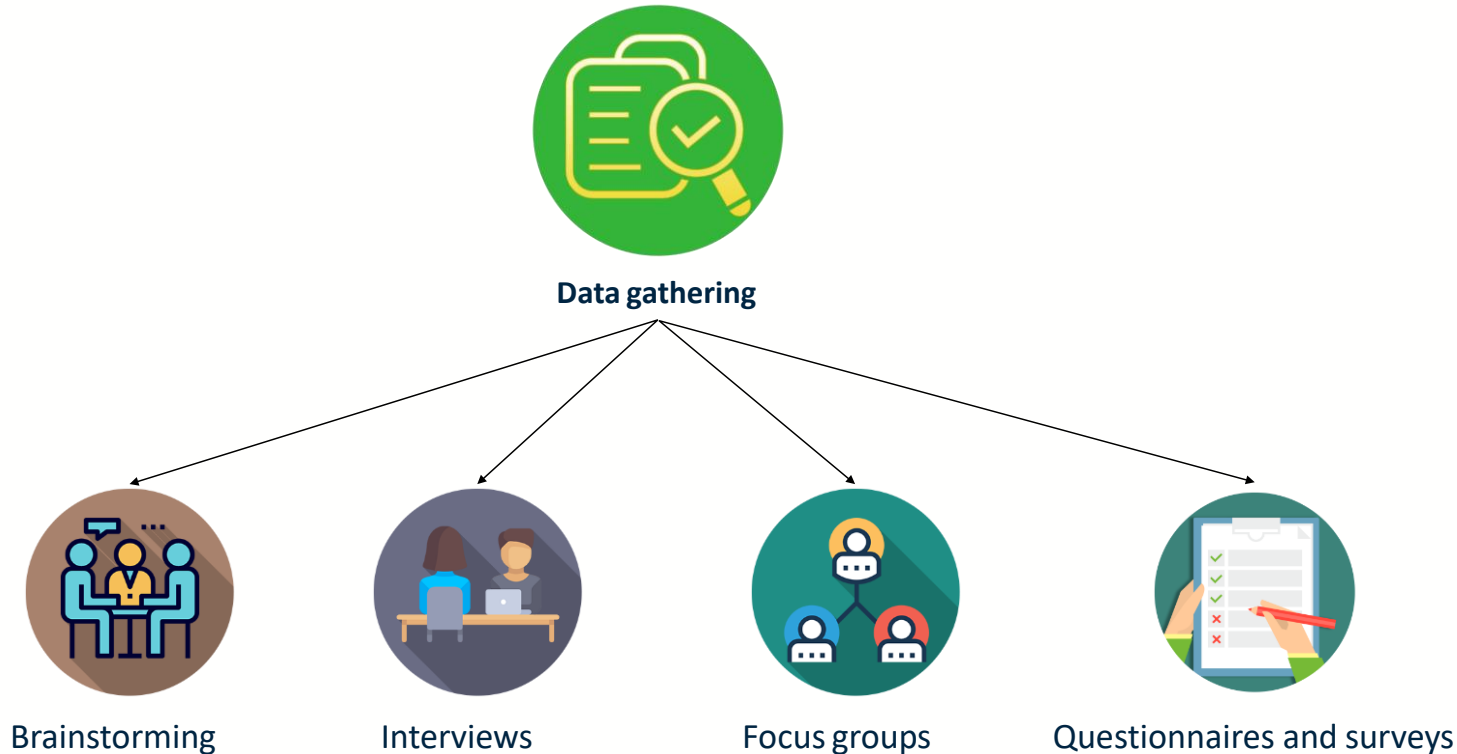
1. Expert Judgment
2. Data analysis
3. Data gathering
4. Decision making
5. Data representation
6. Interpersonal and team skills
7. Context diagram
8. Prototypes



Outputs

1. Requirements documentation
2. Requirements traceability matrix

Collect Requirements - Tools and Techniques



Inputs:

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2. Project management plan
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4. Business documents
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Tools & Techniques:

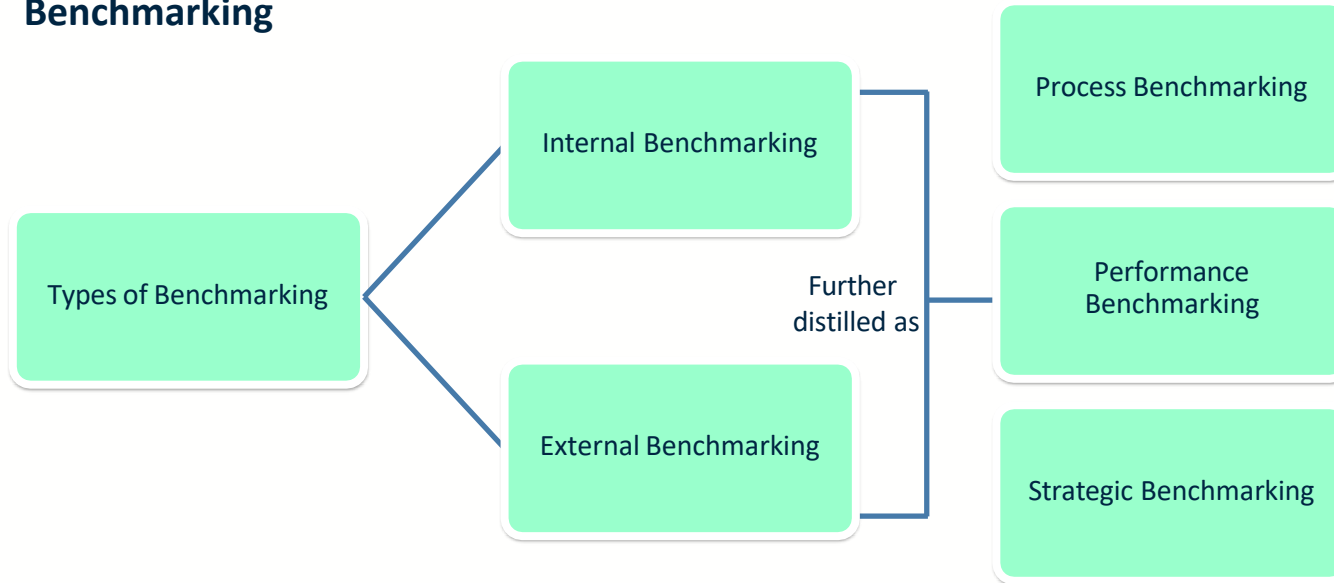
1. Expert Judgment
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Outputs

1. Requirements documentation
2. Requirements traceability matrix

Collect Requirements - Tools and Techniques

Benchmarking



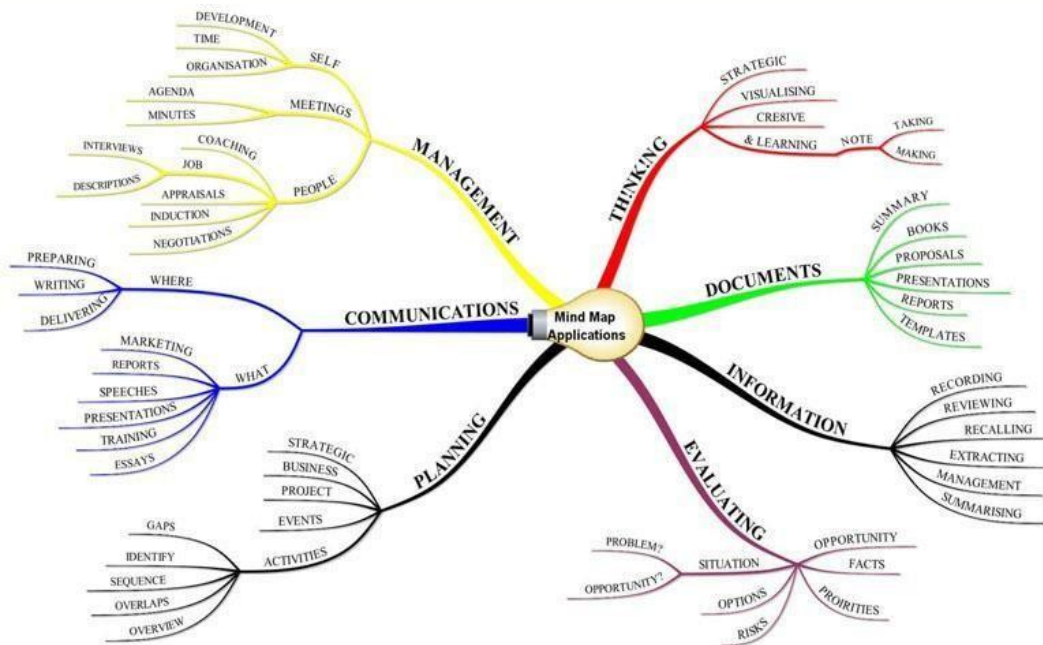
Inputs:	
1.	Project charter
2.	Project management plan
3.	Project documents
4.	Business documents
5.	Agreements
6.	Enterprise environmental factors
7.	Organizational process assets

Tools & Techniques:	
1.	Expert Judgment
2.	Data analysis
3.	Data gathering
4.	Decision making
5.	Data representation
6.	Interpersonal and team skills
7.	Context diagram
8.	Prototypes

Outputs	
1.	Requirements documentation
2.	Requirements traceability matrix

Collect Requirements - Tools and Techniques

- Mind mapping – Consolidated brainstormed ideas are reflected with commonalities and differences to understand and generate new ideas.



Inputs:

1. Project charter
2. Project management plan
3. Project documents
4. Business documents
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7. Organizational process assets

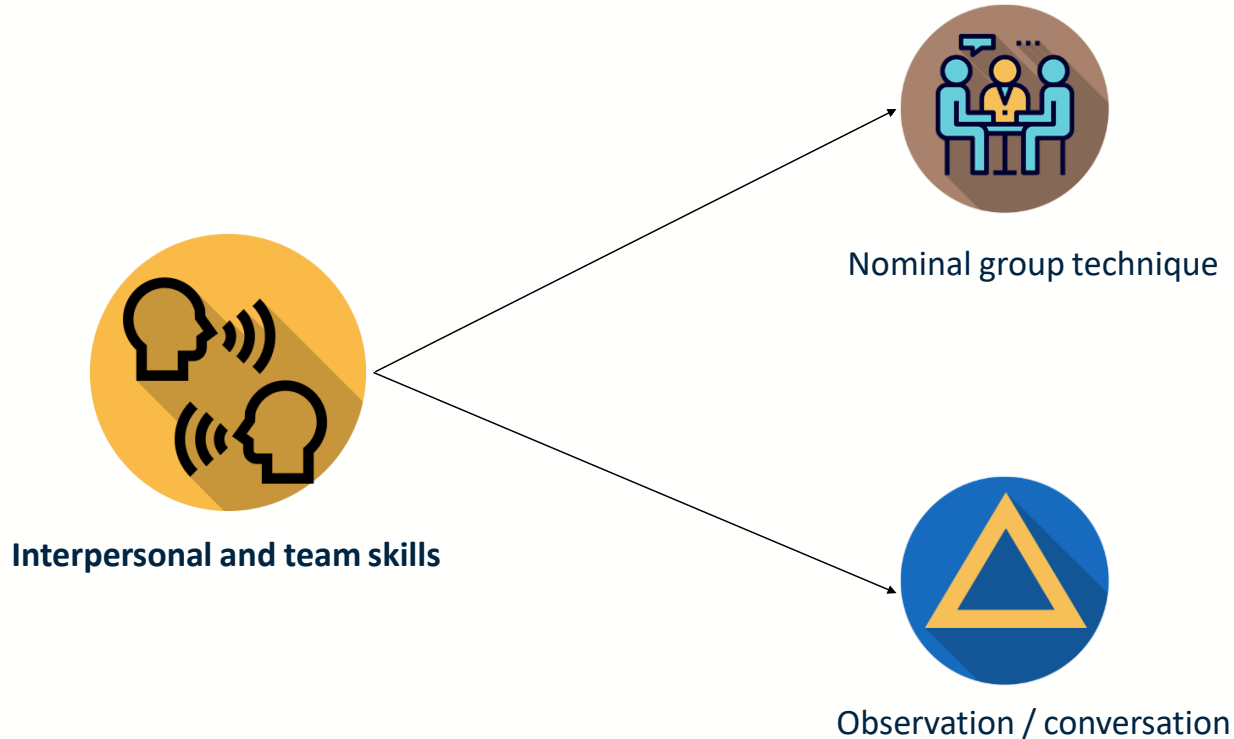
Tools & Techniques:

1. Expert Judgment
2. Data analysis
3. Data gathering
4. Decision making
5. **Data representation**
6. Interpersonal and team skills
7. Context diagram
8. Prototypes

Outputs

1. Requirements documentation
2. Requirements traceability matrix

Collect Requirements - Tools and Techniques



Inputs:

1. Project charter
2. Project management plan
3. Project documents
4. Business documents
5. Agreements
6. Enterprise environmental factors
7. Organizational process assets

Tools & Techniques:

1. Expert Judgment
2. Data analysis
3. Data gathering
4. Decision making
5. Data representation
6. **Interpersonal and team skills**
7. Context diagram
8. Prototypes

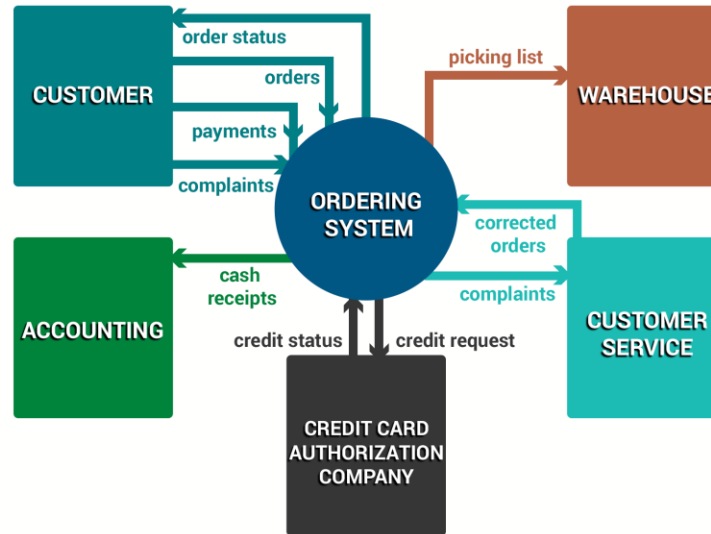
Outputs

1. Requirements documentation
2. Requirements traceability matrix

Collect Requirements - Tools and Techniques

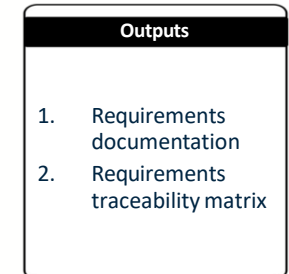
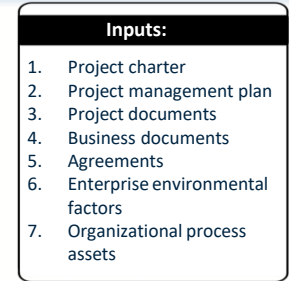
Context diagram

- High level map of a system and its surrounding environment



Prototypes

- A working model of the product before fully developing it. Feedback is given on this to develop it fully.
- Also known as mockups and storyboards depending on industry



Define Scope



- The process of developing a detailed description of the project and product.
- The key benefit of this process is that it describes the product, service, or result boundaries and acceptance criteria



Inputs:

1. Project charter
2. Project management plan
3. Project documents
4. Enterprise Environmental Factors
5. Organizational Process Assets

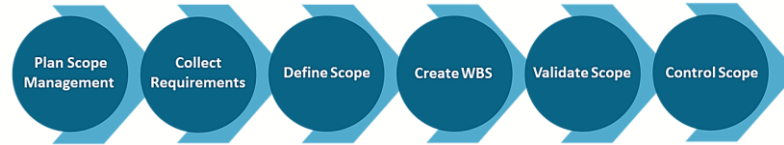
Tools & Techniques:

1. Expert Judgment
2. Data analysis
3. Decision making
4. Interpersonal and team skills
5. Product analysis

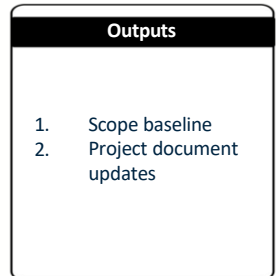
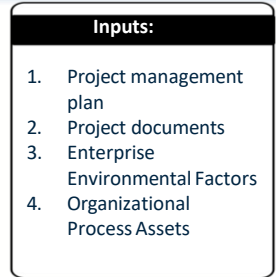
Outputs

1. Project scope statement
2. Project documents updates

Create WBS



- The process of subdividing project deliverables and project work into smaller, more manageable components
- The key benefit of this process is that it provides a framework of what has to be delivered



Create WBS - Tools and Techniques

Expert judgment

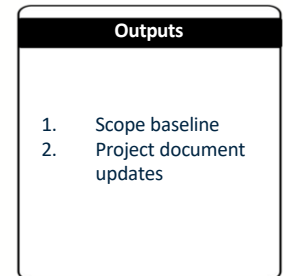
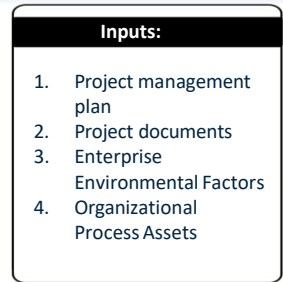
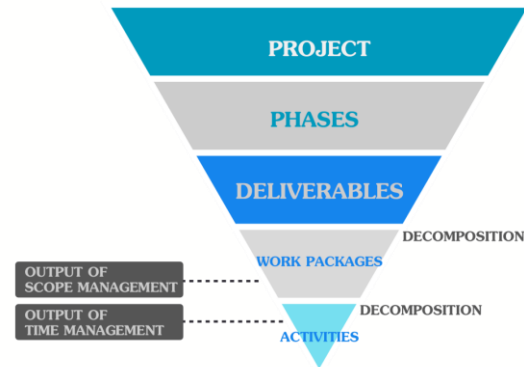
- Expert Judgement is based upon expertise in an application area, knowledge area, or discipline for the appropriate activity being performed.
- Such as related skill, training, specialized education, knowledge, experience etc.



Create WBS - Tools and Techniques

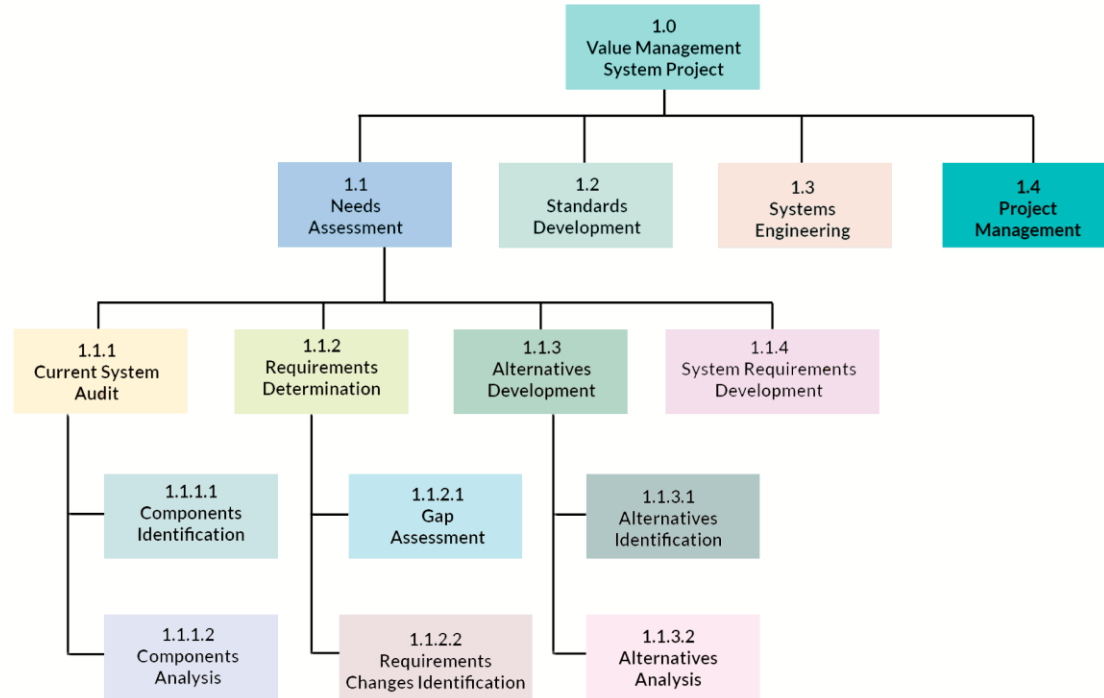
Decomposition

- Identifying and analyzing the deliverables and related work
- Structuring and organizing the WBS
- Decomposing the upper WBS levels into lower-level detailed components
- Developing and assigning identification codes to the WBS components
- Verifying that the degree of decomposition of the deliverables is appropriate



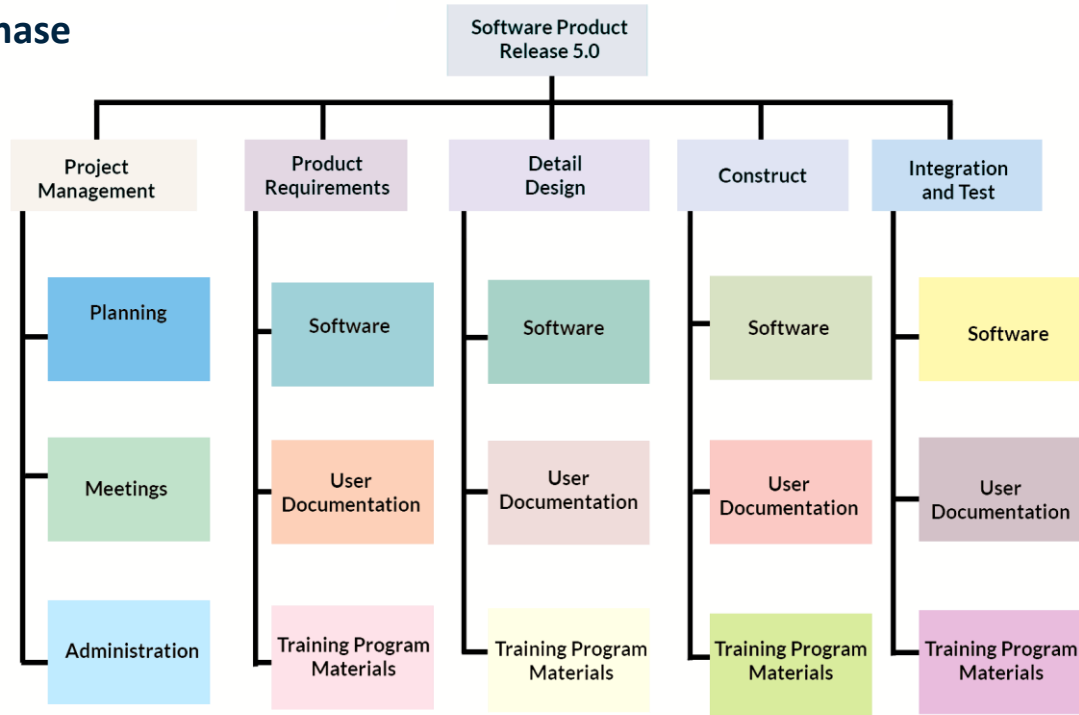
WBS Types

WBS Decomposed Down Through Work Packages



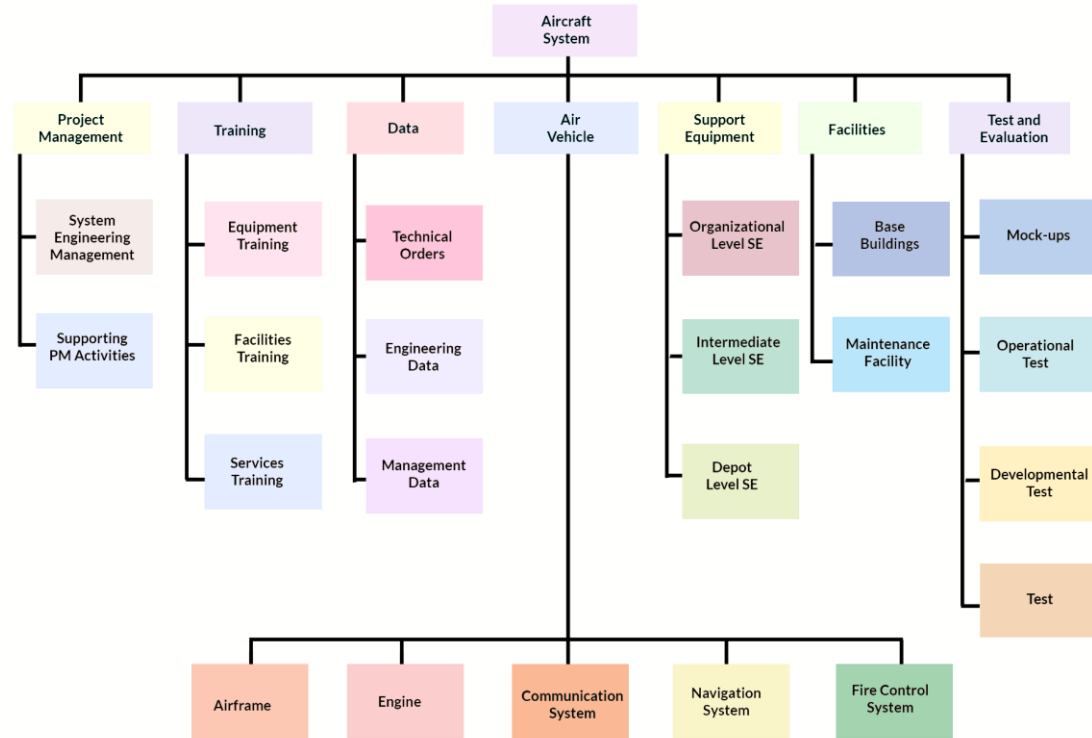
WBS Types

WBS Organized by Phase

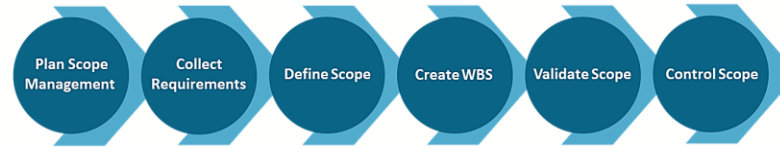


WBS Types

WBS with Major Deliverables



Validate Scope



- The process of formalizing acceptance of the completed project deliverables
- The key benefit of this process is that it brings objectivity to the acceptance process and increases the probability of final product, service, or result acceptance by validating each deliverable



Inputs:

1. Project management plan
2. Project documents
3. Verified deliverables
4. Work performance data

Tools & Techniques:

1. Inspection
2. Decision making

Outputs

1. Accepted deliverables
2. Work performance information
3. Change requests
4. Project documents updates

Validate Scope - Tools and Techniques

Inspection

- Measuring
- Examining
- Validating



Decision making

- Voting



Inputs:

1. Project management plan
2. Project documents
3. Verified deliverables
4. Work performance data



Tools & Techniques:

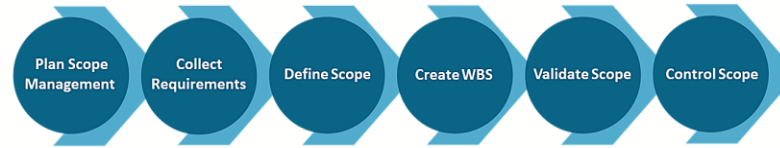
1. Inspection
2. Decision making



Outputs

1. Accepted deliverables
2. Work performance information
3. Change requests
4. Project documents updates

Control Scope



- The process of monitoring the status of the project and product scope and managing changes to the scope baseline
- The key benefit of this process is that the scope baseline is maintained throughout the project



Inputs:

1. Project management plan
2. Project documents
3. Work performance data
4. Organizational process assets

Tools & Techniques:

1. Data analysis

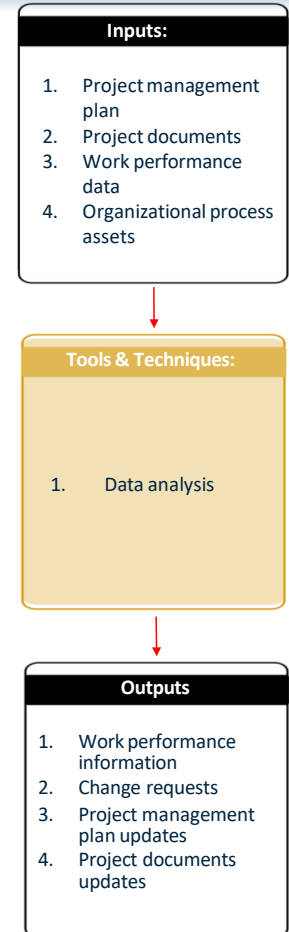
Outputs

1. Work performance information
2. Change requests
3. Project management plan updates
4. Project documents updates

Control Scope - Tools and Techniques

Data analysis

- Variance analysis
 - A technique for determining the cause and degree of difference between the baseline and actual performance.
 - Project performance measurements are used to assess the magnitude of variation from the original scope baseline.
 - Important aspects of project scope control include determining the cause and degree of variance relative to the scope baseline and deciding whether corrective or preventive action is required
- Trend analysis
 - Examines project performance over time to determine if performance is improving or deteriorating



PROJECT SCHEDULE MANAGEMENT



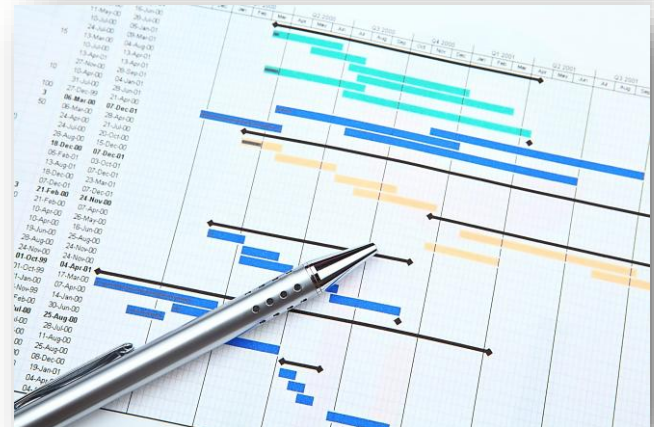
Session Overview

- Project Schedule Management
- Scheduling Overview
- Schedule Management Processes
 - Plan Schedule Management
 - Define Activities
 - Sequence Activities
 - Estimate Activity Durations
 - Develop Schedule
 - Control Schedule



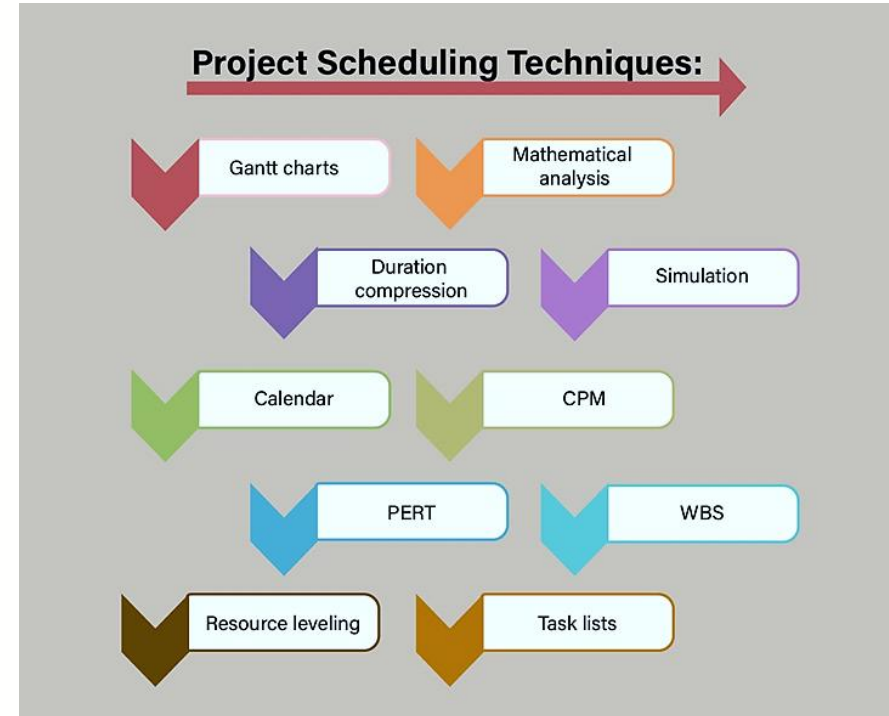
Key concepts

- Creating a schedule model requires **methods** (like **Critical Path Method**), **information** (like **estimates**), and **tools** (like spreadsheets or **software**).



Tailoring considerations

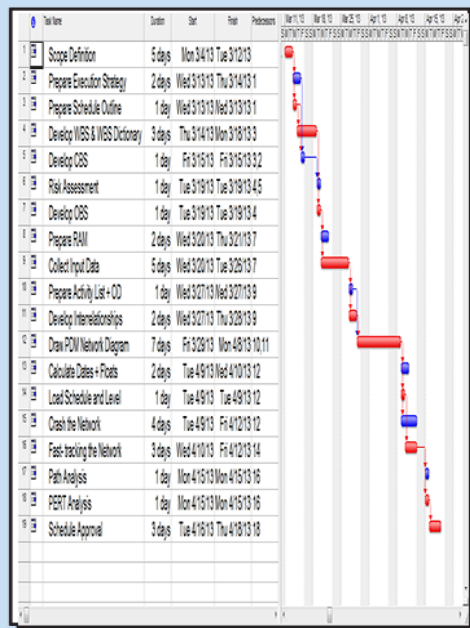
- Lifecycle approach
- Resource availability
- Project dimensions
- Technology support



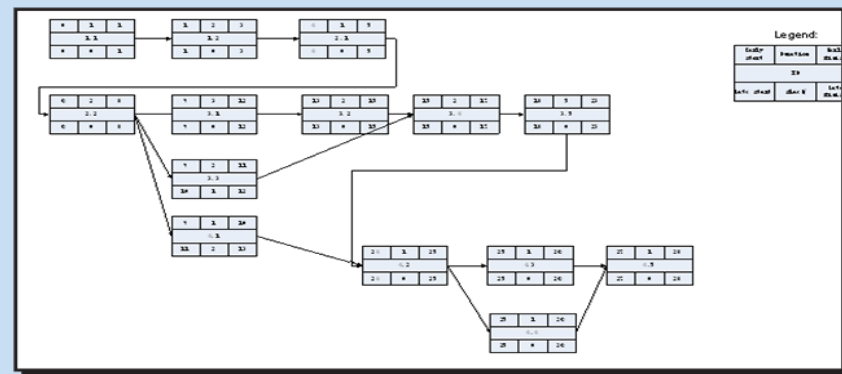
Examples of Project Schedule Presentations

WEEK	DAY	TASK
1	Monday	laundry
	Tuesday	vacuum and dust upstairs and down
	Wednesday	wash bedding touch-up bathrooms
	Thursday	vacuum downstairs kitchen cabinets and appliances
	Friday	laundry
2	Monday	vacuum downstairs clean up clutter
	Tuesday	laundry
	Wednesday	vacuum and dust upstairs and down
	Thursday	wash bedding clean bathrooms
	Friday	vacuum downstairs windows, blinds, & light fixtures
3	Monday	laundry
	Tuesday	vacuum and dust upstairs and down
	Wednesday	wash bedding touch-up bathrooms
	Thursday	vacuum downstairs basement
	Friday	vacuum downstairs clean up clutter
4	Monday	laundry
	Tuesday	vacuum and dust upstairs and down
	Wednesday	wash bedding clean bathrooms
	Thursday	vacuum downstairs kitchen floors
	Friday	laundry

Activity List



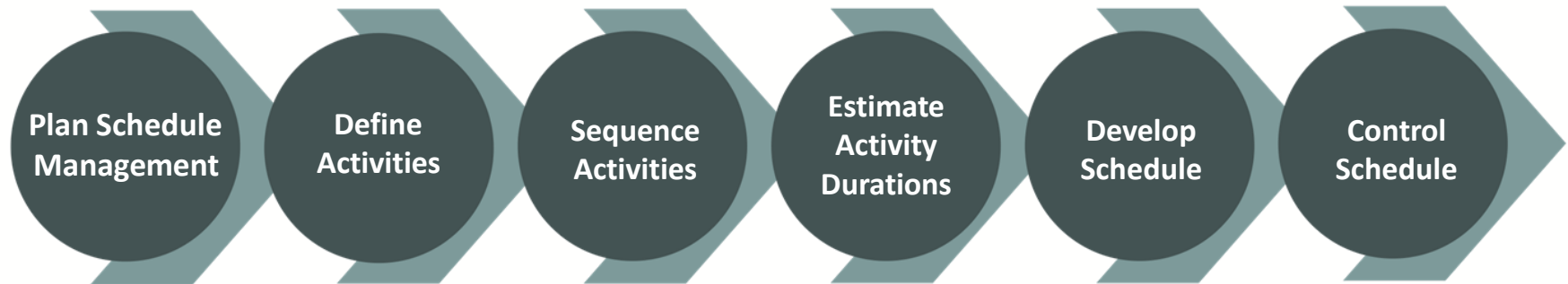
Bar Chart



Network Diagram

Processes in schedule management

- **Project Schedule Management** includes the processes required to manage the **timely completion of the project**.



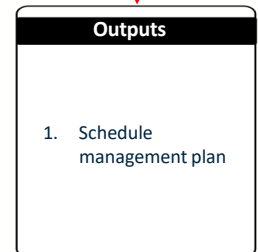
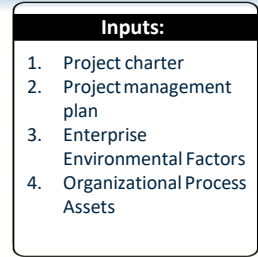
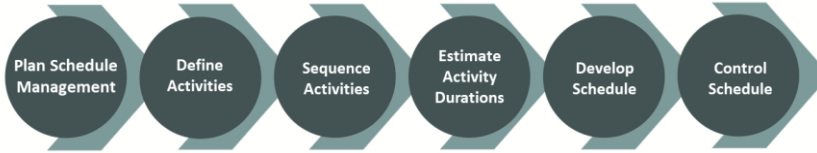
Plan Schedule Management



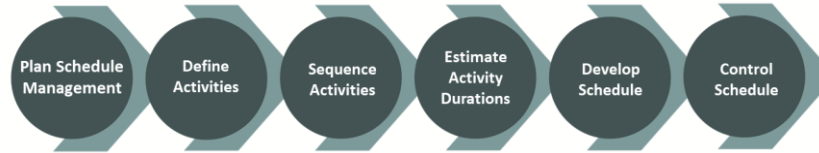
- The process of establishing the policies, procedures, and documentation for **planning, developing, managing, executing, and controlling** the project schedule



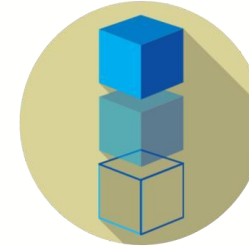
- The key benefit of this process is that it provides guidance and direction on how the project schedule will be managed throughout the project



Define Activities



- The process of identifying and documenting the specific actions to be performed to produce the project deliverables
- The key benefit of this process is that it decomposes its work packages into activities that provide a basis for estimating, scheduling, executing, monitoring, and controlling the project work



Inputs:

1. Project management plan
2. Enterprise environmental factors
3. Organizational Process Assets

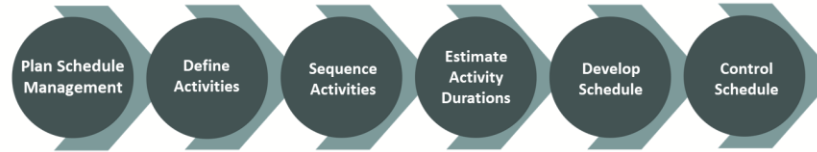
Tools & Techniques:

1. Expert judgment
2. Decomposition
3. Rolling wave planning
4. Meetings

Outputs

1. Activity list
2. Activity attributes
3. Milestone list
4. Change requests
5. Project management plan updates

Sequence Activities



- The process of identifying and documenting relationships among the project activities
- The key benefit of this process is that it defines the logical sequence of work to obtain the greatest efficiency given all project constraints



Inputs:

1. Project management plan
2. Enterprise Environmental Factors
3. Organizational Process Assets

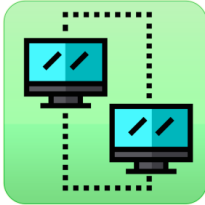
Tools & Techniques:

1. Precedence diagramming method
2. Dependency determination and integration
3. Leads and lags
4. Project management information system

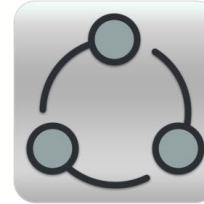
Outputs

1. Project schedule network diagrams
2. Project documents updates

Sequence Activities - Tools and Techniques



Precedence diagramming method (PDM)



Leads and lags



Dependency determination and Integration



Project Information Management System (PMIS)

Inputs:

1. Project management plan
2. Enterprise Environmental Factors
3. Organizational Process Assets

Tools & Techniques:

1. Precedence diagramming method
2. Dependency determination and integration
3. Leads and lags
4. Project management information system

Outputs

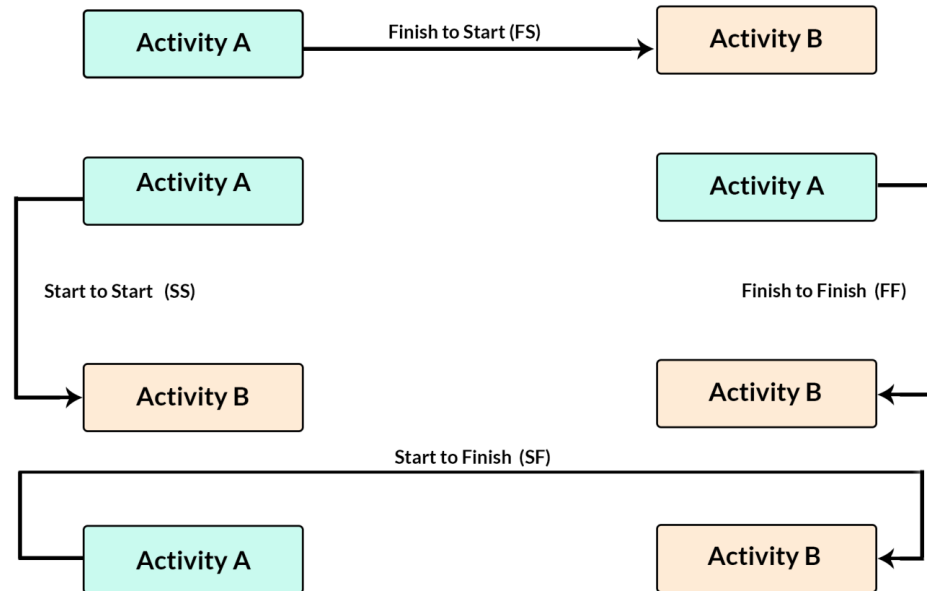
1. Project schedule network diagrams
2. Project documents updates

Precedence diagramming method (PDM)

- A technique used for constructing a schedule model
- Activities are represented by nodes and are graphically linked by one or more logical relationships
- Show the sequence in which the activities are to be performed
- Also known as Activity-on-node (AON): **It's the most common approach to network diagramming**
- A **predecessor activity** is an activity that logically comes before a dependent activity in a schedule
- A **successor activity** is a dependent activity that logically comes after another activity in a schedule

Precedence diagramming method (PDM)

PDM includes four types of dependencies or logical relationships as shown below



Precedence diagramming method (PDM)

Finish-to-start (FS)

Successor activity cannot start until predecessor activity has finished – this is the most common of these



The awards ceremony (successor) cannot start until the race (predecessor) has finished.

Finish-to-finish (FF)

Successor activity cannot finish until predecessor activity has finished



The inspection cannot finish until the painting is completely finished.

Precedence diagramming method (PDM)

Start-to-start (SS)

Successor activity cannot start until predecessor activity has started



Start painting (successor) can begin immediately after you start priming the wall (predecessor).

Start-to-finish (SF)

Successor activity cannot finish until a predecessor activity has started –
this is the least common of these



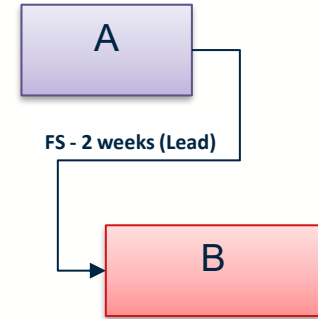
The first security guard shift (successor) cannot finish until the second security guard shift (predecessor) starts

Leads and Lags

Lead is the amount of time whereby a successor activity can be advanced with respect to a predecessor activity. Lead is often represented as a negative value for lag in scheduling software.



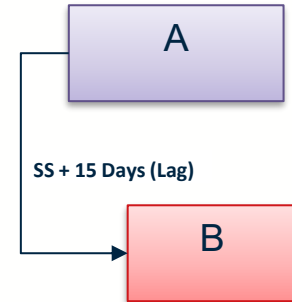
In the project to manufacture a new smartphone, the packaging and preparation for shipment could be scheduled to start two weeks prior to the scheduled completion of the final quality assurance checks and testing.



Lag is the amount of time whereby a successor activity will be delayed with respect to a predecessor activity.



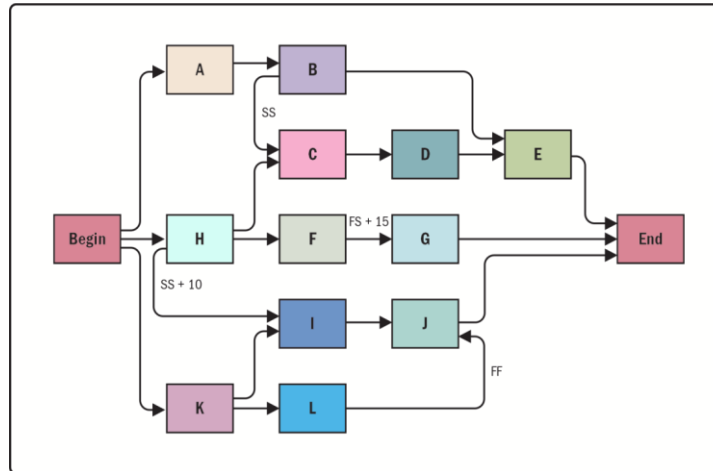
A software development team may commence the debugging and optimization phase of a new software application 15 days after initiating the coding and implementation phase.



Sequence Activities - Outputs

Project Schedule Network Diagrams

- A graphical representation of the logical relationships / dependencies among the project schedule activities.



Inputs:

1. Project management plan
2. Enterprise Environmental Factors
3. Organizational Process Assets

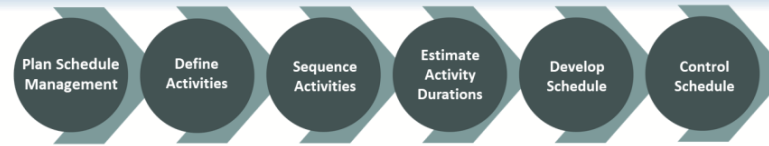
Tools & Techniques:

1. Precedence diagramming method
2. Dependency determination and integration
3. Leads and lags
4. Project management information system

Outputs

1. Project schedule network diagrams
2. Project documents updates

Estimate Activity Durations



- The process of estimating the number of work periods needed to complete individual activities with the estimated resources



- The key benefit of this process is that it provides the amount of time each activity will take to complete



Inputs:

1. Project management plan
2. Project documents
3. Enterprise Environmental Factors
4. Organizational Process Assets

Tools & Techniques:

1. Expert judgment
2. Analogous estimating
3. Parametric estimating
4. Three-point estimating
5. Bottom-up estimating
6. Data analysis
7. Decision making
8. Meetings

Outputs

1. Duration estimates
2. Basis of estimates
3. Project documents updates

Estimate Activity Durations - Tools and Techniques

Three-point estimating

- PERT (program evaluation and review technique) uses three estimates to define an activity's duration - Most likely (M), Optimistic (O), Pessimistic (P)
- Expected duration (E) is calculated in two methods:
 - Triangular Distribution. $E = (O + M + P) / 3$
 - Beta Distribution (from the traditional PERT technique). $E = (O + 4M + P) / 6$

$$\text{Standard deviation } \sigma = \frac{P-O}{6}$$

P – Pessimistic estimate O – Optimistic estimate

$$\text{Variance} = \sigma^2$$



Inputs:

1. Project management plan
2. Project documents
3. Enterprise Environmental Factors
4. Organizational Process Assets

Tools & Techniques:

1. Expert judgment
2. Analogous estimating
3. Parametric estimating
4. Three-point estimating
5. Bottom-up estimating
6. Data analysis
7. Decision making
8. Meetings

Outputs

1. Duration estimates
2. Basis of estimates
3. Project documents updates

Bottom-Up estimating

- Estimating project duration or cost by aggregating the estimates of the lower-level components of the WBS



Example

Q. John is managing a project to build a new project management information system. He works with the team to come up with a most likely estimate of 15 weeks. In the best case, optimistically this could be shortened by two weeks because he can reuse a previous component. But there's a risk that a vendor delay could cause the project to be delayed pessimistically by five weeks. In case using three-point estimating, what are the expected duration based on Triangular Distribution and Beta Distribution separately, and the standard deviation of these estimates?

Answer:

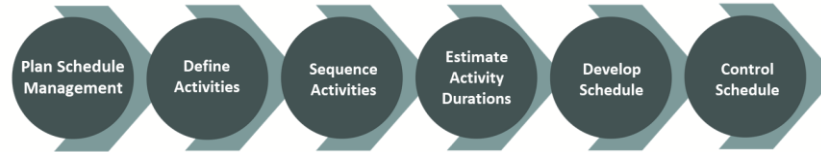
$$O=15-2=13 \quad M=15 \quad P=15+5=20$$

$$\text{Triangular Distribution. } E = (O + M + P) / 3 \quad \rightarrow \quad E = (13 + 15 + 20) / 3 = 16$$

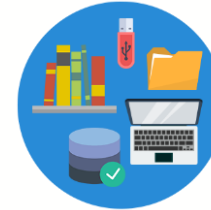
$$\text{Beta Distribution (from the traditional PERT technique). } E = (O + 4M + P) / 6 \quad \rightarrow \quad E = (13 + 60 + 20) / 6 = 15.5$$

$$\text{Standard Deviation} = (P - O) / 6 = (20 - 13) / 6 = 1.17$$

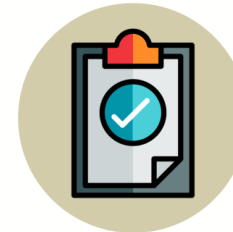
Develop Schedule



- The process of analyzing activity sequences, durations, resource requirements, and schedule constraints to create the project schedule model for project execution and monitoring and controlling



- The key benefit of this process is that it generates a schedule model with planned dates for completing project activities



Inputs:

1. Project management plan
2. Project documents
3. Agreements
4. Enterprise Environmental Factors
5. Organizational Process Assets

Tools & Techniques:

1. Schedule network analysis
2. Critical path method
3. Resource optimization
4. Data analysis
5. Leads and lags
6. Schedule compression
7. Project management information system
8. Agile release planning

Outputs

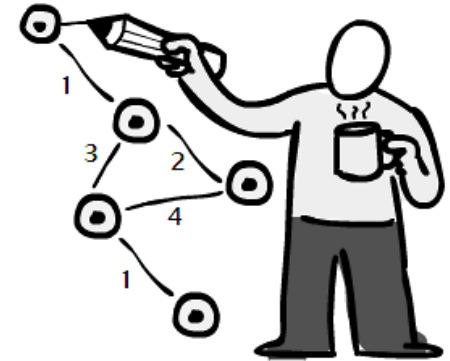
1. Schedule baseline
2. Project schedule
3. Schedule data
4. Project calendars
5. Change requests
6. Project management plan updates
7. Project documents updates

Develop Schedule - Tools and Techniques

Schedule network analysis: Generates the project schedule model using various analytical techniques, e.g., critical path method, resource optimization techniques and modeling techniques.

Critical path method:

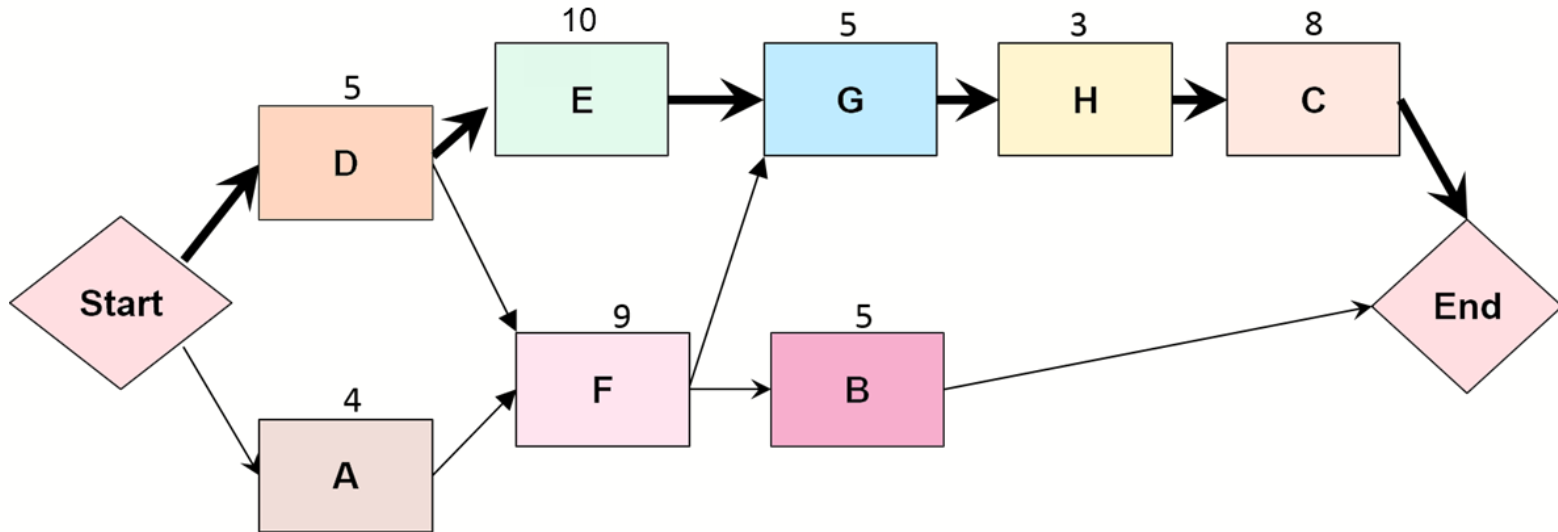
- Critical path is the sequence of activities that represents the **longest path** through a project, which determines **the minimum possible project duration**.
- This is a schedule network analysis technique calculates **the early start, early finish, late start, and late finish dates** for all activities without regard for any resource limitations by performing a **forward and backward pass analysis** through the schedule network.
- **Total Float (Slack)** - the amount of time that a schedule activity can be delayed or extended from its early start date without delaying the project finish date or violating a schedule constraint.



Critical Path Method – Example (1/5)

Activity	Duration	Predecessor
A	4	----
B	5	F
C	8	H
D	5	----
E	10	D
F	9	A,D
G	5	E,F
H	3	G

Critical Path Method – Example (2/5)



Total Number of Paths: 5

1. Start-D-E-G-H-C-End

2. Start-D-F-G-H-C-End

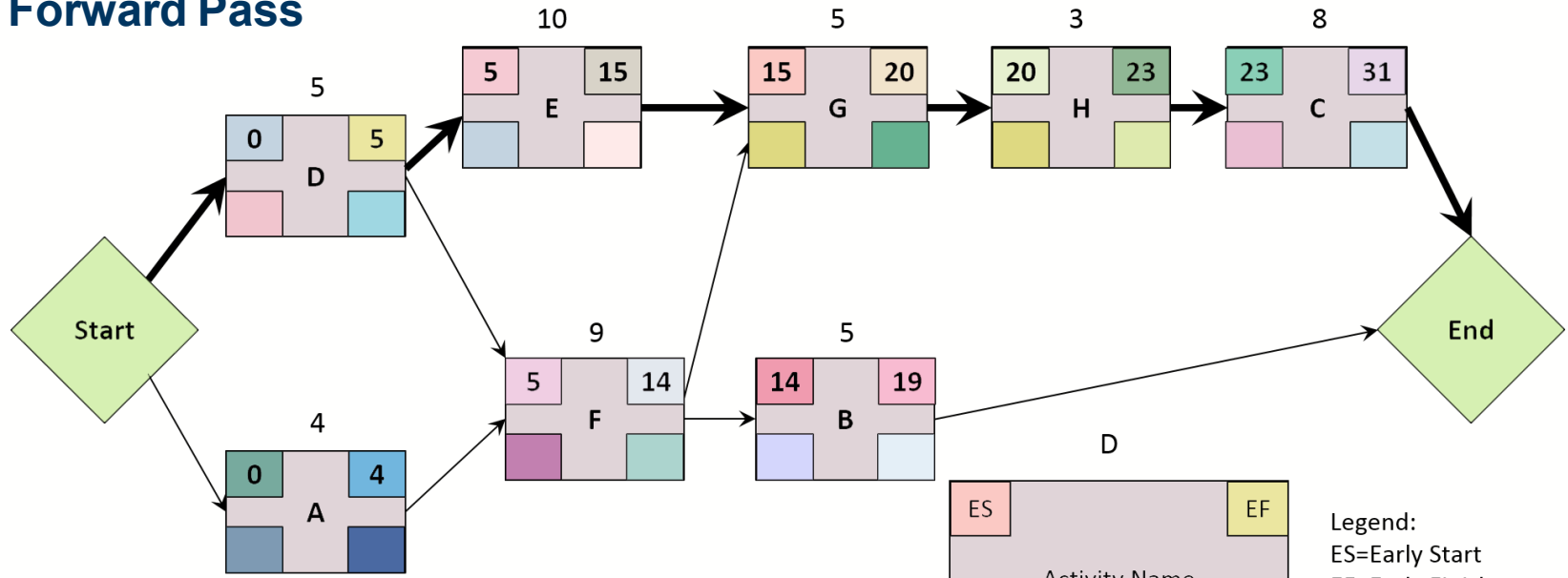
3. Start-D-F-B-End

4. Start-A-F-G-H-C-End

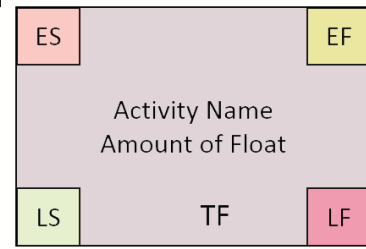
5. Start-A-F-B-End

Critical Path Method – Example (3/5)

Perform Forward Pass



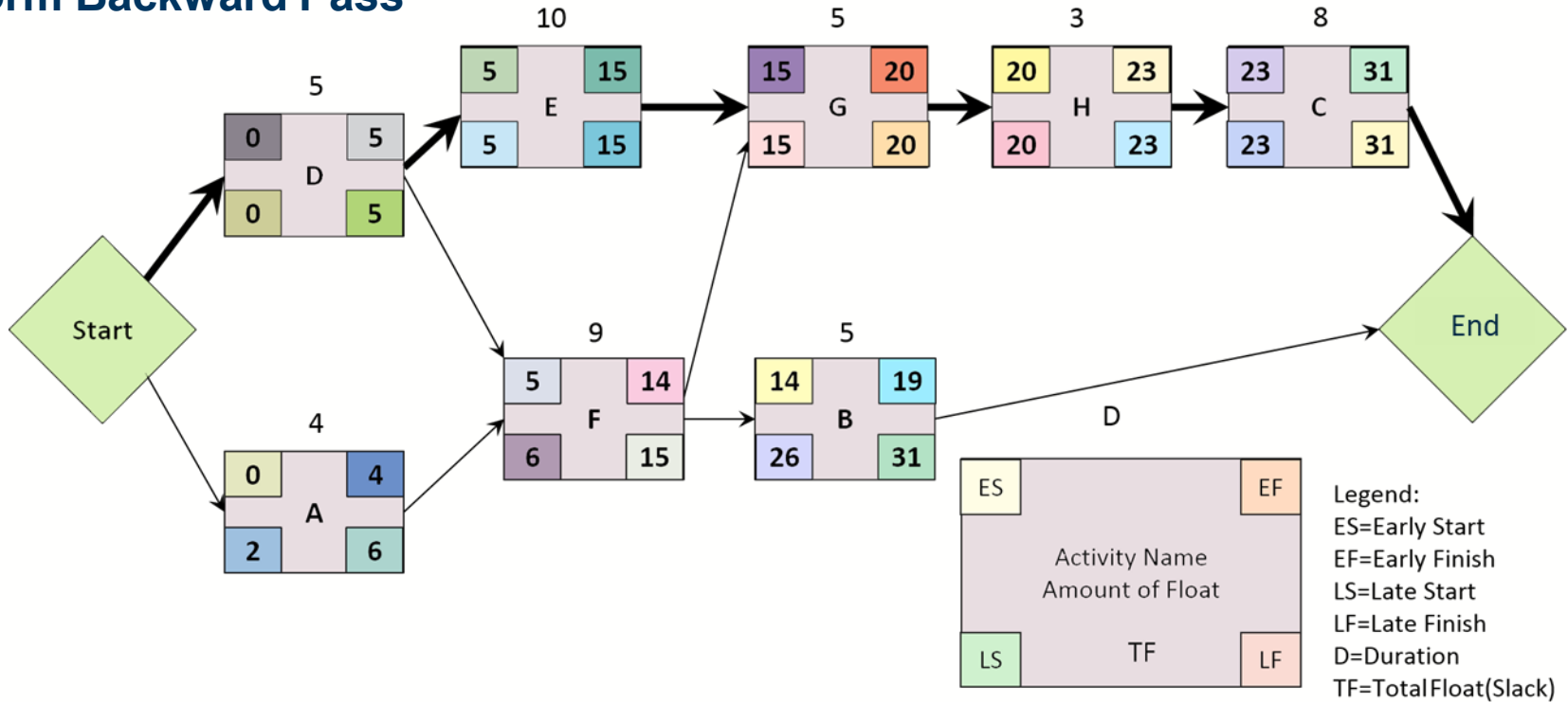
$EF = ES + D - 1$, when we start from 1
 $EF = ES + D$, when we start from 0 (as in this example)



Legend:
 ES=Early Start
 EF=Early Finish
 LS=Late Start
 LF=Late Finish
 D=Duration
 TF=Total Float(Slack)

Critical Path Method – Example (4/5)

Perform Backward Pass

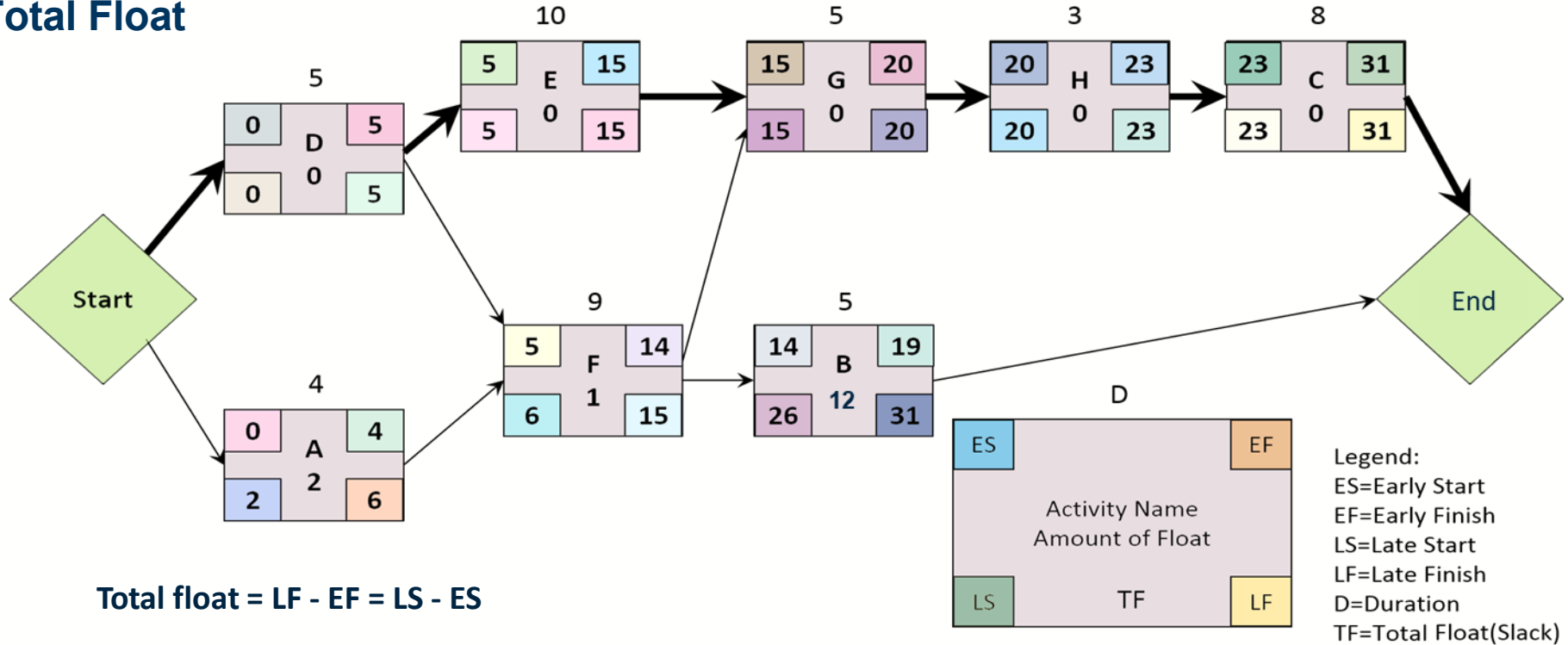


$LF = LS + D - 1$, when we start from 1

$LF = LS + D$, when we start from 0 (as in this example)

Critical Path Method – Example (5/5)

Total Float



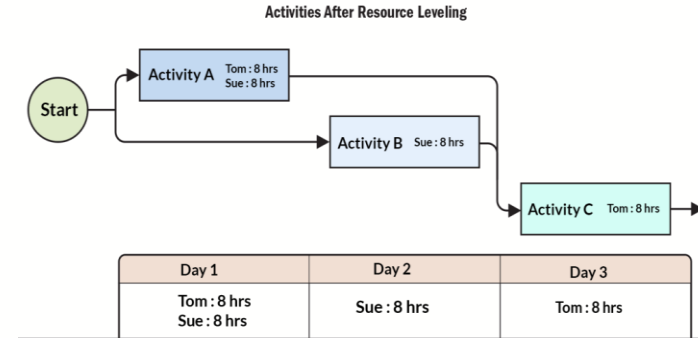
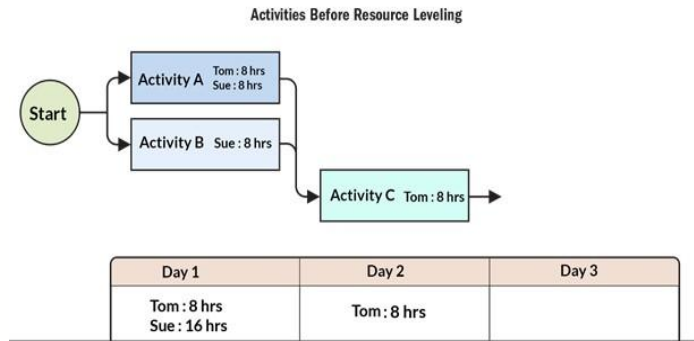
Total float = LF - EF = LS - ES

Critical Path: Start-D-E-G-H-C-End

Duration of Critical Path: 31

Resource Optimization

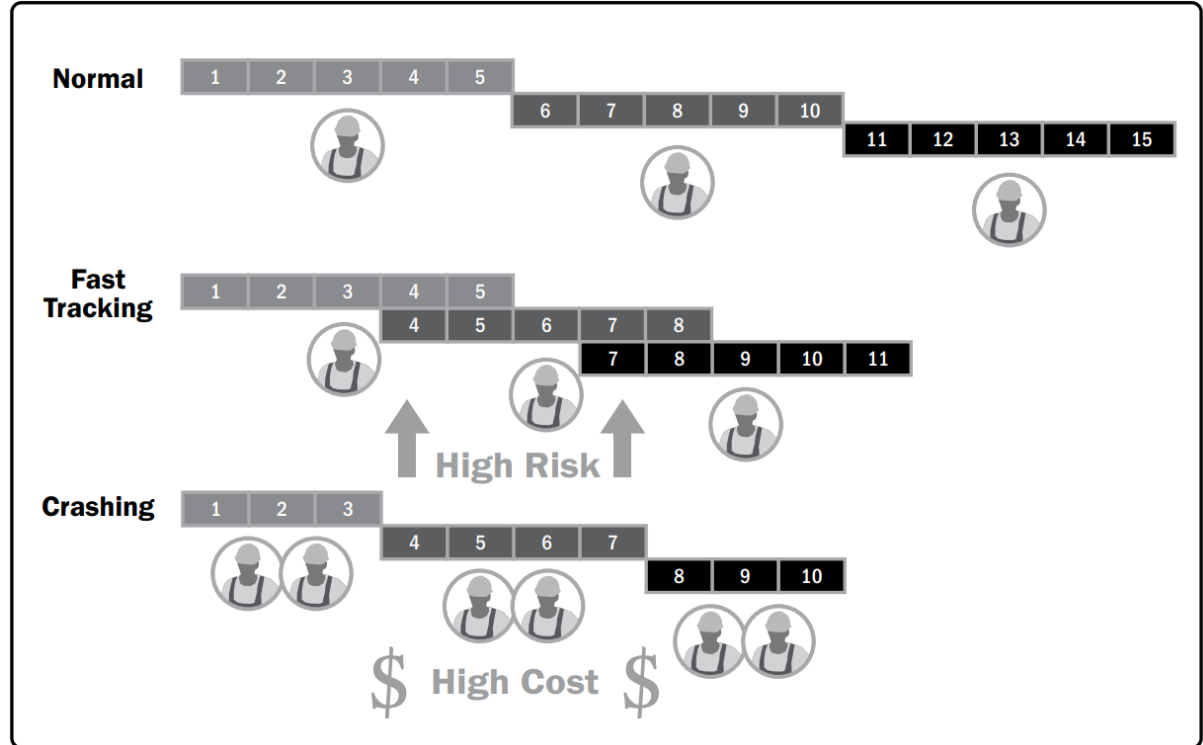
- Start and finish dates are adjusted based on resource constraints with the goal of balancing demand for resources with the available supply (**Resource leveling**)



Schedule Compression

Schedule compression techniques are used to shorten or accelerate the schedule duration without reducing the project scope in order to meet schedule constraints, imposed dates, or other schedule objectives. Main schedule compression techniques are:

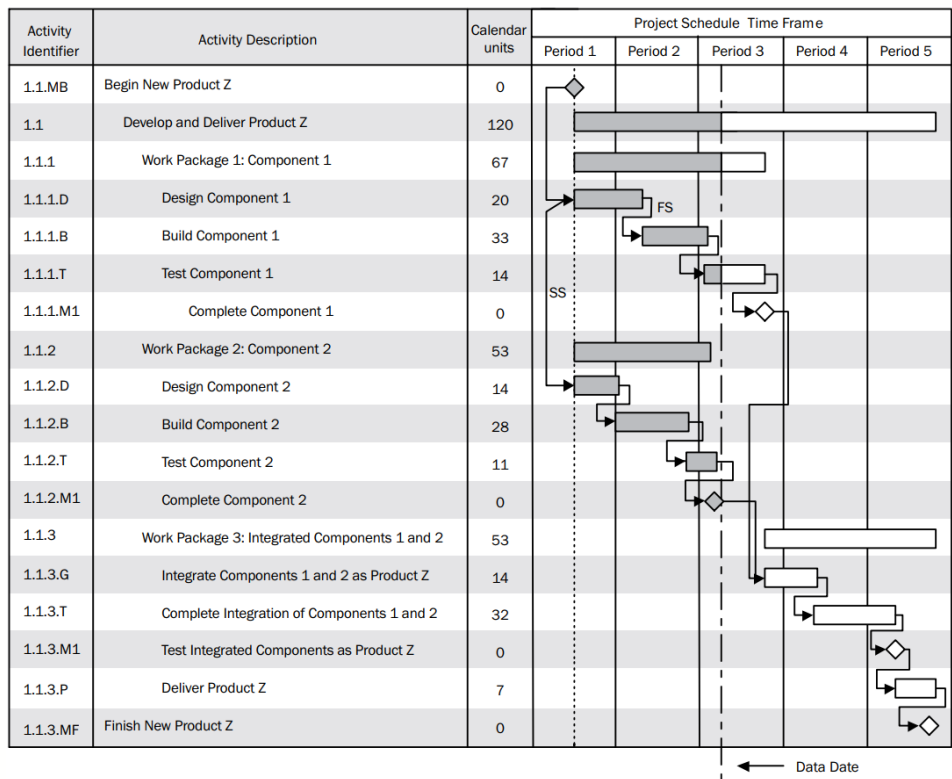
- Crashing
- Fast tracking



Develop Schedule - Outputs

Project schedule

A schedule model that presents linked activities with planned dates, durations, milestones, and resources



Inputs:

1. Project management plan
2. Project documents
3. Agreements
4. Enterprise Environmental Factors
5. Organizational Process Assets

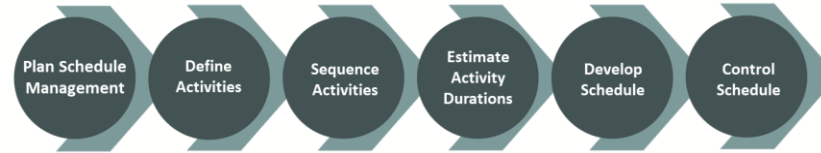
Tools & Techniques:

1. Schedule network analysis
2. Critical path method
3. Resource optimization
4. Data analysis
5. Leads and lags
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8. Agile release planning

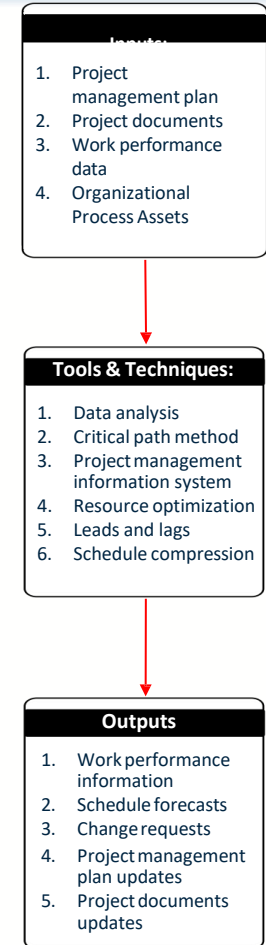
Outputs

1. Schedule baseline
2. **Project schedule**
3. Schedule data
4. Project calendars
5. Change requests
6. Project management plan updates
7. Project documents updates

Control Schedule



- The process of monitoring the status of project to update the project schedule and managing changes to the schedule baseline
- The key benefit of this process is that the schedule baseline is maintained throughout the project







1. Carmen is just about to deliver the case study commissioned by the Tesla Gigafactory project. Before she does, she arranges a meeting with the steering committee and project sponsor associated with the Tesla Gigafactory to take them through the case study, answer any questions or concerns they may have, and seek to get an official confirmation that the case study meets their expectations. What process is she performing here?

- A. Deliverables Acceptance
- B. Accept Deliverables
- C. Close Project or Phase
- D. Validate Scope



2. A Software project manager is working on collecting requirements and establishing estimates for the project. Which process group are we looking at here?

- A. Planning
- B. Executing
- C. Initiating
- D. Monitoring and Controlling



3. Which of the following statements about the WBS is most correct?

- A. WBS stands for Work Breakdown Statement
- B. The WBS includes 100 percent of the work of the project
- C. Any work not included in the WBS is out of scope
- D. B & C



4. Which of the following is the full and correctly ordered list of processes in the Project Schedule Management Knowledge Area?

- A. Define Activities, Sequence Activities, Estimate Activity Durations, Develop Schedule, Control Schedule
- B. Define Activities, Estimate Activity Durations, Develop Schedule, Control Schedule
- C. Plan Schedule Management, Define Activities, Sequence Activities, Estimate Activity Durations, Develop Schedule, Control Schedule
- D. Control Schedule, Sequence Activities, Estimate Activity Durations, Define Activities, Plan Schedule Management, Develop Schedule



5 . Which of the following is true about Resource optimization?

- A.Resource optimization is an input to Estimate Activity Duration, which gives information about available resources to calculate durations
- B.Resource optimization is a tool & technique within Develop Schedule, to adjust resource load levels to make them less variable
- C.Resource optimization is a tool & technique within Estimate Activity Durations, to adjust durations so as to optimize resource load levels
- D.Resource optimization is a tool & technique within Plan Schedule Management, to establish the target rate of resource churn

In the Next Lesson:

Project Management: Part 3

- Project Cost Management
- Project Resource Management

